Few-shot Human Action Recognition with CLIP-based Semantic-Guided Network

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***Abstract*—Given the high cost of manually annotating video data, few-shot action recognition (FSAR) has been proposed to identify novel actions using only a small number of labeled video samples. However, due to the limited availability of annotated data, single-modal initialization still fails to achieve satisfactory performance. In this paper, we aim to leverage the powerful capabilities of CLIP to address the challenges of few-shot action recognition by introducing a CLIP-based text semantic guided few-shot action recognition framework (CSGN). Specifically, we first utilize CLIP image encoder to encode each frame, taking advantage of its strong featureal capacity. We then employ a Transformer to exchange mutual information across frame features. To fully exploit CLIP multimodal properties, we finetuning CLIP visual model using text semantics as a prior. Specifically, we optimize a cross-modal contrastive objective to align video features more closely with their corresponding class-level textual features. Finally, we evaluate CSGN on two widely used few-shot action recognition datasets, comparing it against previous state-of-the-art approaches. The results demonstrate the effectiveness of our framework, highlighting its superiority in leveraging multimodal knowledge for robust action recognition.**

***Keywords*—action recognition, few-shot learning, CLIP**

I. Introduction

In recent years, video has become a primary medium for conveying information in daily life. Video-based technologies, such as video surveillance, content-based video retrieval, and human-computer interaction, have significantly enhanced convenience in various aspects of modern society. In many of these applications, video-based action recognition plays a crucial role. However, training deep learning models for action recognition requires a large amount of carefully curated and labeled data to effectively extract generalizable features. In real-world scenarios, obtaining a substantial volume of labeled video data poses significant challenges. First, manually annotating video data is time-consuming and labor-intensive. Second, in some cases, it may raise concerns regarding user security and privacy. Therefore, this paper focuses on the problem of Few-Shot Action Recognition (FSAR), which aims to identify novel actions using only a limited number of labeled video samples.

Few-shot action recognition remains highly challenging, as directly training a model with a limited amount of labeled data often leads to overfitting and poor generalization. Compared to images, videos introduce an additional temporal dimension, making it even more difficult to extract meaningful features and train a model with strong classification performance. Mainstream few-shot action recognition methods optimize models based on metric learning. These methods first employ a feature extractor to obtain video features and then calculate the distance between query video and support videos using a classification metric. Despite significant progress, the performance of using single-modal remains unsatisfactory due to the limited availability of labeled data.

Recently, Radford et al. introduced CLIP [1], a joint vision-language model trained on 400 million image-text pairs. CLIP demonstrated impressive performance by directly transferring to over 30 datasets without additional fine-tuning. In the field of computer vision, CLIP has been widely adopted as a backbone network for various tasks. These studies have shown that leveraging language assistance can significantly enhance the robustness and generalization ability of the encoder. Inspired by these successful applications, we aim to transfer the powerful capabilities of CLIP to address the challenge of few-shot action recognition. Specifically, we utilize CLIP's pre-trained visual encoder to encode each frame, leveraging its powerful feature capability. We then employ a Transformer to exchange mutual information across frame features. To

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fully exploit CLIP's multimodal properties, we propose a novel CLIP-based text-semantic-guided framework for few-shot action recognition. To adapt CLIP to the few-shot video setting, we optimize a video-text contrastive objective, aligning video features more closely with their corresponding class-level textual features. Additionally, we refine the visual model within CLIP using text semantics as a prior. To assess the impact of each component on few-shot action recognition, we conduct detailed ablation studies and benchmark our approach against previous state-of-the-art methods on two widely used few-shot action recognition datasets. The results demonstrate that CSGN outperforms existing approaches, highlighting its effectiveness in leveraging multimodal information for robust few-shot action recognition.

II. Related work

A. Few-Shot Learning (FSL)

Few-Shot Learning aims to correctly recognize novel classes with only a few labeled samples. Few-Shot Learning first emerged in the field of computer vision and has received extensive attention in recent years. Few-Shot Learning methods can be categorized into three types based on their approaches. Data augmentation-based methods utilize auxiliary information to alleviate the scarcity of labeled data in target domains. Optimization-based approaches aim to develop models that can learn quickly and effectively adapt to new classes. On the other hand, metric-based approaches focus on learning to minimize the distance between support features and query features. For instance, Matching networks [2] proposed matching networks to learn a metric space and then determine the category of query samples based on the distance between query and support samples. Prototypical networks [3] introduced prototype networks, where each class prototype is represented by averaging its support samples. They then compare query samples with each prototype using Euclidean distance.

B. Few-Shot Action Recognition (FSAR)

Few-Shot Action Recognition is a branch of Few-Shot Learning, which aims to recognize novel actions with only a few labeled videos. Compared with image recognition, the extra temporal dimension of actions makes classification more challenging. Most existing FSAR methods are mainly based on metric learning. For example, TRX [4] proposed a novel approach to few-shot action recognition, finding temporally-corresponding frame tuples between the query and videos in the support set. AmeFu-Net [6] alleviates the problem of extreme data scarcity by introducing depth information as the carrier of the scene. Wang et al. [7] proposed a method including a long-short contrastive objective and a motion autodecoder to address the problem that the matching process between local frames is often inaccurate due to the lack of guidance to enforce long-range temporal perception.

III. Methodology

A. Problem formulation

The task of few-shot action recognition aims to accurately classify an unlabeled query video into one of several categories, where these categories have not been seen during training. Specifically, there is a training set $\mathcal{D}_{train} = \{(\mathrm{x}_i, l_i), l_i \in \mathcal{C}_{train}\}$ and a test set $\mathcal{D}_{test} = \{(\mathrm{x}_j, l_j), l_j \in \mathcal{C}_{test}\}$, where x_i and x_j represent videos from the training and test sets, respectively, and l_i and $l_j$denote their corresponding labels. The categories in the training set $\mathcal{C}_{train}$ are completely disjoint from those in the test set $\mathcal{C}_{test}$, meaning there is no category overlap between them. The model is trained on $\mathcal{D}_{train}$ and then evaluated on $\mathcal{D}_{test}$, where it must recognize novel categories with only one or a few labeled examples per class.

Since each new category has only a limited number of labeled videos, we employ episodic training to learn more effective feature features. Each episode consists of a support set and a query set. In a C-way K-shot setting, each episode is constructed by sampling C classes, with K labeled videos per class in the support set. The query set is then composed of additional samples drawn from the remaining videos of these C classes. The objective is to leverage the $C \times K$ labeled support videos to accurately classify each query video into one of the C classes.

B. Preliminaries of CLIP

Contrastive Language-Image Pretraining (CLIP) leverages a contrastive loss function to align image and text features, enabling the model to directly comprehend visual content described in natural language. Specifically, CLIP consists of a visual encoder and a text encoder. The visual encoder maps input images into a feature space, while the text encoder extracts high-level semantic features from natural language descriptions. The model is trained such that matching image-text pairs are pulled closer together in the shared embedding space, while non-matching pairs are pushed apart. CLIP is pre-trained on 400 million image-text pairs collected from the internet, allowing it to develop strong multimodal feature capabilities.

C. Cross-Modal Contrastive Learning

In this paper, we represent the set of support videos and query video as $X = \{x_i\}_1^N$ and $x_q \in \mathbb{R}^{L \times H \times W \times C}$, respectively, where L is the length of the video. We utilize CLIP's image encoder to extract frame-level features, as shown in Figure 1. To capture temporal dependencies within the video, we employ a Transformer to exchange mutual information across frames. These two encoders together serve as our video feature extractor $\mathcal{V}(\cdot)$ by

$$f_i^v = \mathcal{V}(x_i) \tag{1}$$

Furthermore, we freeze the text encoder $\mathcal{V}(\cdot)$ and extract textual features using a prompt template, "a video of [CLS]". The obtained text features are represented as $\{f_i^t\}_1^{|\mathcal{D}_{train}|}$.

To bridge the gap between the text encoder and the video encoder, we first employ contrastive learning to minimize the distance between video features and their corresponding class-level textual features while maximizing the separation between features of different classes. This approach enables the video

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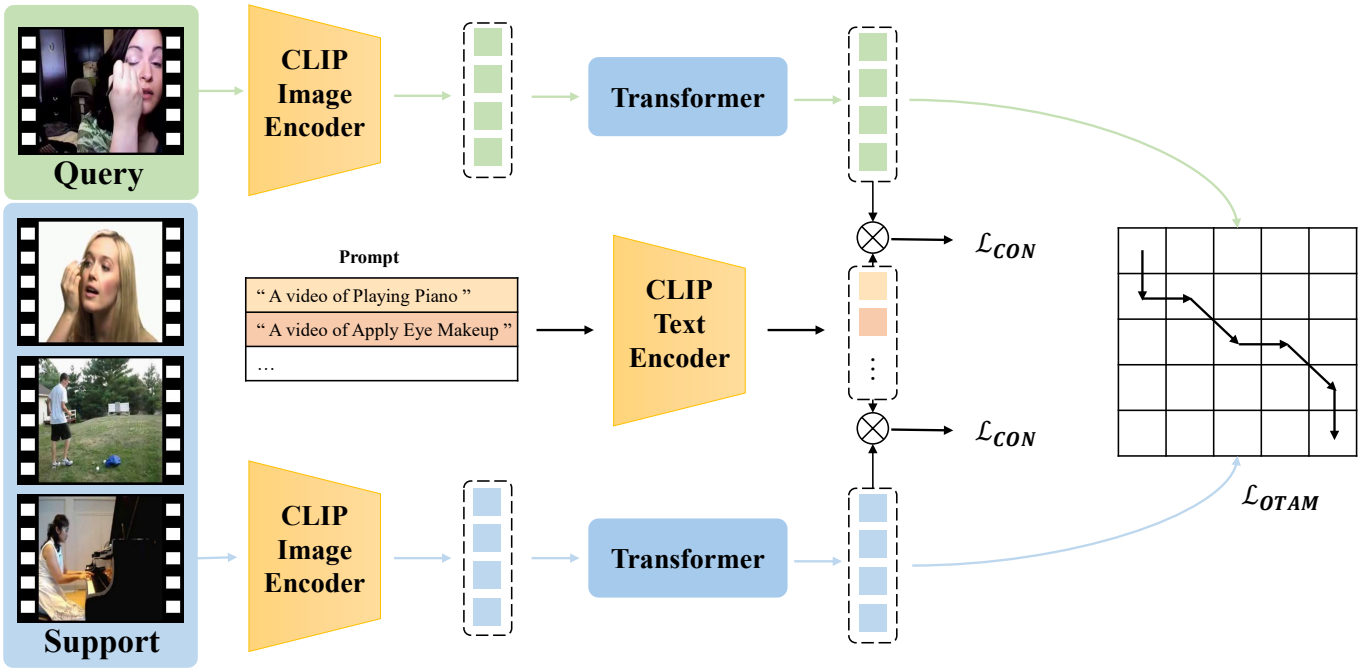


Fig. 1. The framework of CSGN.

encoder to learn a feature space that is well-aligned with the text encoder's feature space. Thus, we compute the cross-Modal contrastive loss as follows:

$$\mathcal{L}_{CON} = -\sum_{i}^{N} \log \frac{\exp(Sim(Pooling(f_i^v), f_i^t)/\tau)}{\sum_{j=1}^{|\mathcal{D}_{train}|} \exp(Sim(Pooling(f_i^v), f_j^t)/\tau)} \tag{2}$$

where $Sim(\cdot,\cdot)$ is the similarity measurement (i.e., cosine similarity) and $\tau \in \mathbb{R}$ is a hyper-parameter representing the weight.

D. Few-shot Classifier with OTAM

Existing few-shot action recognition methods typically classify a query video x_q by computing its metric distance to support video prototypes. The metric distance between the query video x_q and a support video x_i can be expressed as:

$$D_{q,i} = \mathcal{M}(f_q^v, f_i^v) \tag{3}$$

where $\mathcal{M}(\cdot,\cdot)$ denotes the metric function used for distance computation. In Ordered Temporal Alignment Module (OTAM), the metric distance is learned by integrating segment-wise distances along an ordered temporal alignment path, ensuring that the predicted distance preserves temporal order. Based on the OTAM metric distance, the probability distribution of the query video q over the support classes can be formulated as:

$$\mathcal{L}_{OTAM} = -\log \frac{\exp(-\mathcal{M}(f_q^v, f_p^v))}{\sum_j \exp(-\mathcal{M}(f_q^v, f_j^v))} \tag{4}$$

where f_p^v is the class prototype of the query video x_q. In this way, we design the final loss for a batch of query videos as follows.

$$\mathcal{L} = \mathcal{L}_{OTAM} + \lambda \mathcal{L}_{CON} \tag{5}$$

In the above formula, $\mathcal{L}_{OTAM}$ enables the model to learn to recognize action, and $\mathcal{L}_{CON}$ enables the model to further enhance feature extraction for the video encoder. The hyper-parameter $\lambda \in (0, 1)$ is used to strike a balance between the two sub-sets during training.

IV. Experiments

A. Dataset

We conduct experiments on two commonly used few-shot action recognition datasets: UCF101 and HMDB51.

UCF101 [8]: UCF101 includes 101 action classes with 13, 320 video clips. In our work, we follow the partitioning of [6], i.e., we use 70 action classes as training set, 10 action classes as validation set, and 21 action classes as test set.

HMDB51 [9]: HMDB51 includes 51 action classes with 6, 849 video clips. Each action class has at least 101 video clips. Following the work of [7], 31/10/10 non-overlapping classes are sampled to train, validate and test.

B. Implementation Details

Data Sampling For video pre-processing, we first sample 1 frame per 4 frames from the video. We adopt the same sampling strategy as [7], which sparsely samples 8 frames from each of video clips and resizes them to 256×256 pixels. Then we randomly crop the resized frames into the size of 224 × 224.

Training details: In our work, we conduct the experiments under the 5-way K-shot setting, where K = 1, 3, 5. For each

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TABLE I
AVERAGE ACCURACY OF 5-WAY FEW-SHOT ACTION RECOGNITION ON UCF101 AND HMDB51 AND DATASETS.

Method	Backbone	UCF101			HMDB51		
		1-shot	3-shot	5-shot	1-shot	3-shot	5-shot
TRX [4]	ResNet50	78.2	92.4	96.1	53.1	66.8	75.6
OTAM [5]	ResNet50	79.9	87.0	88.9	54.4	65.7	68.0
HCL [11]	ResNet50	82.5	91.0	93.9	59.1	71.2	76.3
HyRSM [12]	ResNet50	83.9	93.0	94.7	60.3	71.7	76.0
MTFAN [10]	ResNet50	84.8	-	95.1	59.0	-	74.6
AMeFu-Net [6]	ResNet50	85.1	93.1	95.5	60.2	71.5	75.5
GgHM [13]	ResNet50	85.2	-	**96.3**	61.2	-	76.9
MoLo [7]	ResNet50	86.0	93.4	95.5	60.8	72.0	**77.4**
CSGN	ResNet50	**87.2**	**93.5**	95.2	**62.8**	**72.5**	76.3

episode, we randomly select K samples for each of 5 action classes. We use the ResNet50 as our backbone network to extract the feature of each frame. Following the contrastive learning literature [14], the temperature parameter in the contrastive loss is set to 0.07 (i.e., τ=0.07). The hyperparameter λ for balancing the training of labeled and unlabeled data in Eq. 5 is set to 0.6 by default.

We train the model for 10,000 episodes using Adam optimizer. The learning rate of our model l_r is set to 1×10^{-4}, and l_r is reduced by 0.7 after 400 episodes.

We implement the proposed method using Python and Pytorch. The experiments were conducted on a computer equipped with Intel(R) Xeon(R) Silver 4210R CPU @ 2.40GHz, two NVIDIA GeForce RTX 3090 24G graphics card.

Testing details. During testing, in order to make a fair comparison, we only use the encoder $\mathcal{V}(\cdot)$ as the testing model to get video-level features, while removing other components used in training. We evaluate CSGN on 5-way 1-shot, 5-way 3-shot and 5-way 5-shot benchmarks. Then, we report average results over 10,000 episodes randomly sampled during testing.

C. Comparison with State-of-the-Arts

We compare CSGN with existing work on two commonly used FSAR datasets. For a fair comparison, we use ResNet50 as the backbone network and report accuracy using 8 frames. As shown in Table I, CSGN outperforms all state-of-the-art methods in both 1-shot and 3-shot settings while achieving competitive results in the 5-shot setting. On the UCF101 dataset, CSGN achieves accuracy rates of 87.2%, 93.3%, and 95.5% for the 1-shot, 3-shot, and 5-shot settings, respectively. Similarly, on the HMDB51 dataset, CSGN attains 62.8%, 72.5%, and 76.3% under the same settings. Compared to OTAM, our approach demonstrates significant performance gains of 7.3% (87.2% vs. 79.9%) on UCF101 and 8.4% (62.8% vs. 54.4%) on HMDB51 in the 1-shot setting. These results highlight the effectiveness of CSGN in leveraging CLIP's multimodal knowledge to substantially enhance few-shot action classification performance.

D. Ablation Study

To further validate the effectiveness of introducing CLIP, we consider three typical baseline methods. The first is OTAM with a ResNet-50 backbone pre-trained on ImageNet, which is trained end-to-end for few-shot action recognition. The second replaces the ImageNet-pretrained ResNet-50 with CLIP's visual encoder, where the encoder's parameters are frozen, and it is used solely for feature extraction while applying OTAM's metric learning for classification. The third builds upon the second by fine-tuning CLIP's visual encoder to further adapt it to the few-shot action recognition task. As shown in Table II, CLIP's visual encoder already achieves competitive results without fine-tuning. Moreover, fine-tuning CLIP on the few-shot action recognition task allows it to better adapt to downstream tasks, leading to further performance improvements.

TABLE II
IMPACT OF DIFFERENT TRAINING STRATEGIES. AVERAGE ACCURACY OF 5-WAY 1-SHOT FSAR ON UCF101 DATASET IS REPORTED.

Pre-train	Training	UCF101 5-way 1-shot
ImageNet	Finetuning	80.3
CLIP	Freezed	83.2
CLIP	Finetuning	**87.2**

E. Hyperparameter Analysis

Influence of the Number of Transformer Blocks: The number of Transformer blocks in the video encoder is an important hyperparameter in the proposed model. We evaluate the performance with different numbers of Transformer blocks. As shown in Table III. It shows that as the number of Transformer blocks increases, the performance of the model increases at the beginning. After reaching some number, the performance degrades when the blocks increases. The reason may be that noise in the input data makes it harder for deeper networks to generalize, and more parameters in the model may lead to overfitting. Based on the average accuracy, we consider 3 Transformer blocks to be suitable for the proposed model.

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TABLE III
IMPACT OF DIFFERENT NUMBER OF TRANSFORMER BLOCK. AVERAGE ACCURACY OF 5-WAY 1-SHOT FSAR ON UCF101 DATASET IS REPORTED.

	UCF101 5-way 1-shot
Transformer Block × 1	83.5
Transformer Block × 2	86.1
Transformer Block × 3	**87.2**
Transformer Block × 4	86.3

TABLE IV
IMPACT OF HYPERPARAMETER λ. AVERAGE ACCURACY OF 5-WAY 1-SHOT FSAR ON UCF101 AND HMDB IS REPORTED.

λ	UCF101 5-way 1-shot	HMDB51 5-way 1-shot
0.2	81.5	55.6
0.4	85.5	57.3
0.6	**87.2**	**62.8**
0.8	86.9	62.3

Influence of Hyperparameter λ: In the poposed contrastive learning approach, we use Equation 5 to integrate supervised and unsupervised learning, where the hyperparameter λ is used to balance their weight. As shown in Table IV, when λ is small, the model will focus more on few-shot action recognition and cannot effectively utilize the multi-modal information of CLIP. When λ is large, the model will focus more on the supervision information of the text modality and cannot effectively extract the modal information of the video itself. Based on the accuracy, we consider that a λ value of 0.6 helps to balance the model training.

V. CONCLUSION

In this paper, we aim to leverage the powerful capabilities of CLIP to address the challenges of few-shot action recognition by introducing a CLIP-based text semantic guided few-shot action recognition framework. Specifically, we first utilize CLIP image encoder to encode each frame, taking advantage of its strong featureal capacity. We then employ a Transformer to exchange mutual information across frame features. To fully exploit CLIP multimodal properties, we finetuning CLIP visual model using text semantics as a prior. Specifically, we optimize a cross-modal contrastive objective to align video features more closely with their corresponding class-level textual features. Finally, we evaluate CSGN on two widely used few-shot action recognition datasets.

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